Power BI Project: Electronics Sales Data

Objective:

This project is designed to guide you through the entire **Power BI workflow** for a real-world scenario involving **dirty sales data**. You'll be tasked with cleaning, analyzing, and modeling the data to create a comprehensive **dashboard** and a **detailed report** that includes key business insights. The dataset includes various **real-world challenges** such as **inconsistent data**, **missing values**, and **currency discrepancies**. Your goal is to transform this data into something useful for decision-making in an **electronics retail business**.

Instructions:

Step 1: Data Cleaning

Data cleaning is often the most time-consuming part of any analytics project. It's essential to ensure that your data is both **accurate** and **consistent**. Below are a few things to consider:

1. **Ensure Consistency in Data Types**:
   Some columns might have inconsistent data types. For example, the `OrderDate` column should be a **date**, and `SalesAmount` should be a **numeric** type. Check for any **text** values in these columns that don't match the expected format.
2. **Check for Spelling Errors**:
   You'll likely find some inconsistencies in **product names**, **country names**, or even **customer emails**. For example, "Samsung Galaxy S21" might have variations like "Samsung Galaxy S21" and "Samsung GAlaxy S21" in the dataset. Tip: Filter for product names to help standardize them.
3. **Missing or Incomplete Data**:
   Look for **empty cells** or missing values in columns such as `Profit`, `SalesAmount`, and `ShippingCost`. It's important to decide whether to **remove**, **impute**, or **fill** these gaps.
4. **Pseudonyms and Invalid Entries**:
   Sometimes, customers or sales representatives might be recorded as "Anonymous" or "N/A." These entries need to be cleaned or handled to avoid inconsistencies in analysis.

   For example:

   - A row with `CustomerName = "N/A"` in a paid sales transaction should be flagged for review and ensure all pseudoblanks are consistent if any.
5. **Ensure Correct Data Formats**:
   Ensure that columns such as **dates** (OrderDate, DeliveryDate) and **currency** (USD, EUR) are consistently formatted. You might need to standardize date formats across the dataset.

   Example:

- If `Currency` has values like `"USD"`, `"usd"`, `"US Dollar"`, unify them to a single standard format, e.g., `"USD"`.

Step 2: Data Analysis

Once the data is cleaned, it's time to dive into the **analysis**. Your goal is to extract meaningful insights and calculate key business metrics using **DAX** (Data Analysis Expressions) to create calculated measures.

1. **Calculate Total Sales**:
   What is the **total revenue** from all products? You should create a **measure** that sums up `SalesAmount` for all rows in your cleaned dataset.
2. **Profit Margin Calculation**:
   Profit margin is critical for understanding the profitability of your business. The formula to calculate profit margin is:
   Profit Margin=profit/SalesAmount ×100
   You need to create a **measure** that calculates this for each transaction and possibly by **product**, **region**, or **time period**.
3. **Research Market Rates for Currency Conversion**:
   The dataset uses different currencies (USD, EUR, GBP, etc.). You will need to research the **current market exchange rates** for the currencies used in the dataset. Using this data, **create a DAX measure** that converts **all currencies to USD** (or whichever currency you prefer). The **rates** can be obtained from various online sources.

   Example:

   - If you find the current exchange rate of **EUR to USD** is **1.1**, use this to convert all **EUR sales** into **USD**.
4. **Handling Missing or Incorrect Values**:
   What happens when some rows have missing data, or when certain sales figures are too low (e.g., a negative profit)? How will you treat these cases in your analysis?

   Question to consider:

   - Should you exclude transactions with missing or extreme values, or should you adjust these values based on your business knowledge?

Step 3: Data Modeling

Modeling in Power BI is about structuring your data efficiently, so that it can be easily analyzed and visualized.

1. **Create Dimension Tables**:
   You'll need to create **dimension tables** to categorize and group your data. Think of dimension tables as lookup tables that store information like **Customer Information**, City, **Product Categories**, **Sales Reps**, and **Time**.

Example:

- **Product Dimension**: Contains columns for `ProductName`, `Category`. This will help in analyzing which products perform the best.

2. **Fact Table**:
   The **fact table** will store transactional data such as `OrderID`, `SalesAmount`, `Profit`, `Quantity`, and `Discount`. It's important to link your fact table to the dimension tables using **relationships**.

   Question to consider:

   - How will the fact table relate to the **customer**, **product**, and **time** tables? Create the appropriate **relationships** in Power BI.

3. **Time Dimension**:
   A **Time Dimension** is crucial for time-based analysis. This table should include columns for `Year`, `Quarter`, `Month`, `Week`, `Day`, and `Date`. This will allow you to create **time-based aggregations** and trends (e.g., monthly sales, yearly growth).

Step 4: Building the Dashboard

Your goal here is to create an interactive **Power BI dashboard** that allows stakeholders to gain valuable insights from the data.

1. **Key Visualizations** examples: use only what you deem necessary or important for your dashboard.
   Consider the following visualizations that can bring out meaningful insights:
   - **KPI Cards**: Display key metrics such as **Total Sales**, **Total Profit**, and **Profit Margin** etc
   - **Bar Chart**/ Column- Show **sales by product**, **sales by region**, or **sales by category**.
   - **Line Chart**: Display **sales trends over time** (monthly, quarterly).
   - **Pie Chart**: Show the **sales distribution by currency**, or **sales by region**.
   - **Map Visual**
2. **Slicers**:
   Use **slicers** to allow users to filter the dashboard by **region**, **customer**, **time period**, and **product category**. This will make your dashboard interactive.

   Example:

   - A slicer for **year** will allow users to analyze sales data for different years.

Step 5: Creating the Report

Once your dashboard is built, create a **detailed report** with all the necessary insights. Here are the key components to include:

1. **Executive Summary**:
   Provide a brief summary of key insights, such as overall sales performance, top-performing products, and profitability.
2. **Detailed Analysis**:
   Include in-depth analysis for specific metrics. For example, provide a detailed breakdown of **total sales by region**, **profit margins by product**, and **yearly trends**.
3. **Visual Insights**:
   Use charts, tables, and slicers to highlight trends and make the report interactive. Ensure the report answers questions like:
   - What are the top-performing products?
   - Which region has the highest profit margin?
   - How has sales been trending over the last year?
4. **Recommendations**:
   Based on the data, provide business recommendations, such as:
   - Which products should be promoted based on high sales and profit margins?
   - How should the company address regions with lower profitability?

Example questions to Nudge Your Analysis:

- How do **exchange rate fluctuations** affect your sales figures? How will you account for this in your model?
- What happens if a customer makes multiple purchases? Should you consider **repeat business** in your analysis?
- Is it better to **exclude rows** with missing values, or should you try to **fill them** using **historical data**?

Power BI reports are created within the **Power BI Desktop** application and consist of multiple **pages** or **tabs**, each containing different visualizations and insights.

How the Report is Structured in Power BI:

1. **Pages in a Power BI Report**:
   - **Power BI Reports** can have multiple **pages** (just like slides in a PowerPoint presentation).
   - Each **page** in the report can be dedicated to a specific part of the analysis. For example, one page could be focused on **Sales by Product**, another on **Profit Margins**, and another on **Geographical Insights**.
   - To add a new page, click on the **"+" (New Page)** button at the bottom of the Power BI Desktop window. You can have as many pages as needed for your report.
2. **The Main Page (Dashboard)**:

    - The first page (the default page) of your report is typically where you'll place your **KPI cards**, **interactive slicers**, and **main visualizations**. This acts as a **dashboard** for the report, providing high-level insights at a glance.
    - This is where users will likely interact with the data (e.g., filter by region, product, etc.).
3. **Additional Pages**:
    - Subsequent pages can focus on more detailed analysis. For example:
        - **Page 1**: High-level KPIs and sales trends.
        - **Page 2**: Product-specific sales and profitability.
        - **Page 3**: Geographical sales analysis by region or store.
        - **Page 4**: Detailed customer or store performance analysis.
4. **Page Navigation**:
    - Power BI allows you to customize how users navigate between pages by adding **buttons** or **links** within the report, such as "Next Page" or "Back to Dashboard."

Steps to Create and Organize Pages in Your Report:

1. **Create a New Page**:
    - Click on the **"+"** icon at the bottom of Power BI Desktop to create a new page.
    - You can right-click on the page tab and rename it to something meaningful, like "Sales Overview," "Product Performance," or "Geographical Insights."
2. **Design the First Page (Dashboard)**:
    - Add **KPI Cards** for key metrics like **Total Sales**, **Total Profit**, **Profit Margin**, and any other relevant metrics you want to display at a glance.
    - Include **bar charts**, **pie charts**, or **line graphs** that show trends over time (e.g., monthly or yearly performance).
    - Use **slicers** to allow users to filter data by key attributes like **Product**, **Region**, or **Store**.
3. **Create Additional Pages**:
    - After creating your first page, click the **"+"** icon to add new pages.
    - For each new page, focus on one specific analysis, such as **Sales by Region** or **Profitability Analysis**.
4. **Interactivity Between Pages**:
    - To allow users to navigate between pages, you can add **buttons** or **links** that can be set to navigate to other pages.
    - For example, on a **store performance page**, you can add a **button** that allows the user to drill down to more detailed product data.
5. **Final Report Structure**:
    - Typically, the **first page** acts as a high-level overview or dashboard.
    - Additional pages can dive deeper into specific areas, such as **products**, **customers**, **sales trends**, and **store performance**.

Navigation Between Pages:

- **Power BI Navigation**: Users can simply click on the page tabs at the bottom to move between pages in the report.

- You can also add **buttons** for navigation (e.g., "Go to Next Page"), which allows users to navigate seamlessly across pages in the report.

Each **page** in your Power BI report serves a specific purpose, from high-level overview to detailed analysis. The **main page** of your report could serve as an interactive dashboard that users can filter and interact with, while other pages dive deeper into specific analyses, such as **product performance**, **geographical analysis**, and **store insights**.

Conclusion:

This project challenges you to apply the full **Power BI workflow**, from **data cleaning** to **modeling** and **dashboard creation**. By the end of this project, you will have built a comprehensive Power BI report and dashboard that provides valuable insights into the **electronics sales business**.

Your ability to clean data, use **DAX**, **model data**, and create **interactive visualizations** will be critical in delivering a functional and insightful report.